

HUICHON, Korea, Democratic People's Republic ofWMO: **470390**Lat: **40.167N**Lon: **126.250E**Elev: **509**StdP: **14.43**Time Zone: **9.00 (E09)**Period: **90-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	1.0	-8.9	3.5	0.4	-5.0	4.3	3.4	12.3	19.2	10.0	18.3	0.4	320	0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	89.1	77.3	86.7	75.7	84.4	74.2	80.1	86.8	78.0	84.2	76.3	81.8	3.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.0	148.4	85.2	76.1	139.1	82.6	74.6	131.8	80.4	44.2	86.7	42.0	84.2	40.3	81.9	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	9.5	7.7	-8.6	94.1	5.3	2.1	-12.4	95.6	-15.5	96.9	-18.5	98.0	-22.4	99.5		
(5)			-8.7	82.9	5.1	2.6	-12.4	84.8	-15.4	86.3	-18.3	87.8	-22.0	89.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.0	17.4	24.4	36.4	49.1	60.9	69.9	75.6	75.6	65.8	52.4	36.5	22.0
	DBStd	21.30	7.81	7.53	6.85	6.89	5.37	4.33	3.82	4.25	5.51	6.67	8.76	8.14
	HDD50	3578	1010	716	423	97	2	0	0	0	0	53	409	868
	HDD65	6768	1475	1136	886	477	149	10	0	1	56	391	854	1333
	CDD50	3203	0	0	2	71	339	596	793	793	476	128	5	0
	CDD65	918	0	0	0	1	21	156	328	329	82	1	0	0
	CDH74	6713	0	0	0	26	324	1119	2286	2432	510	16	0	0
	CDH80	1909	0	0	0	2	58	285	699	791	73	1	0	0
Wind	WSAvg	1.6	1.2	1.7	2.1	2.6	2.4	1.8	1.5	1.2	1.2	1.4	1.3	1.1
Precipitation	PrecAvg	44.9	0.4	0.8	1.1	2.2	3.3	4.8	13.3	10.0	4.3	2.1	1.9	0.9
	PrecMax	77.3	1.3	3.8	3.5	3.8	6.1	10.6	40.0	41.9	18.3	5.1	4.8	2.3
	PrecMin	26.1	0.0	0.1	0.0	0.2	0.7	1.7	3.7	2.0	0.5	0.1	0.4	0.1
	PrecStd	13.2	0.3	0.9	0.7	1.0	1.5	2.2	7.5	7.6	3.6	1.2	1.2	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	49.1	62.9	77.6	85.6	89.9	93.3	92.9	85.1	76.2	61.8	42.7
		MCWB	34.4	41.4	50.3	60.9	67.9	71.6	81.1	80.4	72.6	63.0	53.3	37.9
	2%	DB	35.1	43.6	57.3	72.4	80.9	86.1	89.0	89.4	82.2	71.5	56.5	39.2
		MCWB	31.4	37.1	46.9	57.9	65.5	71.1	78.2	78.6	70.3	60.4	50.2	36.3
	5%	DB	32.5	40.0	52.6	68.0	77.5	83.3	86.5	86.9	79.6	68.0	52.8	36.2
		MCWB	29.3	34.8	44.3	54.9	63.5	70.4	77.2	76.5	68.5	58.0	48.0	33.3
	10%	DB	29.7	36.6	48.6	63.6	74.0	80.5	83.9	84.5	76.9	64.4	49.3	33.7
		MCWB	26.8	32.3	41.8	52.5	61.6	69.3	75.9	75.2	67.0	55.8	45.3	31.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.4	42.6	52.6	62.8	70.7	76.2	84.1	84.5	76.1	65.0	54.5	40.3
		MCDB	39.1	48.6	60.9	74.3	81.2	83.7	90.9	90.3	82.1	73.2	59.5	41.3
	2%	WB	32.1	37.9	48.0	59.0	67.0	73.7	80.4	80.4	72.8	61.8	51.2	36.4
		MCDB	34.5	42.2	55.2	70.3	77.8	82.6	87.1	87.4	78.8	68.9	55.4	38.7
	5%	WB	29.7	35.2	45.0	56.1	64.7	72.2	78.3	78.4	70.6	59.5	48.4	33.7
		MCDB	32.2	39.4	52.0	66.4	75.3	80.5	84.4	84.8	77.2	65.9	52.5	35.7
	10%	WB	26.9	32.7	42.2	53.1	62.7	70.6	76.7	76.6	68.6	57.1	45.6	31.6
		MCDB	29.6	36.2	48.0	61.8	72.1	78.1	82.1	82.3	74.8	62.9	48.8	33.6
Mean Daily Temperature Range	5% DB	MDBR	20.7	19.9	19.0	22.0	21.8	18.7	13.9	16.2	20.4	22.5	16.9	17.3
		MCDBR	20.6	21.8	25.9	30.0	29.2	24.7	19.1	19.8	23.1	27.0	21.4	16.5
		MCWBR	17.0	15.8	17.1	16.9	15.5	12.5	11.1	11.0	13.4	16.2	15.3	13.4
	5% WB	MCDBR	18.9	19.7	24.1	27.4	25.7	19.7	15.7	16.0	19.5	23.4	19.4	14.9
		MCWBR	16.0	14.7	16.7	16.9	15.1	12.0	10.7	10.4	12.4	15.2	14.6	12.6
Clear-Sky Solar Irradiance	taub		0.379	0.456	0.527	0.602	0.627	0.688	0.645	0.556	0.486	0.510	0.452	0.391
	taud		1.982	1.840	1.740	1.656	1.649	1.599	1.742	1.945	2.051	1.876	1.933	1.993
	Ebn at Noon		232	229	227	220	219	206	214	230	238	210	204	215
	Edh at Noon		40	54	66	77	79	83	71	57	48	51	42	37
All-Sky Solar Radiation	RadAvg		803	1070	1381	1547	1745	1618	1352	1402	1373	1067	708	654
	RadStd		41	72	103	125	148	164	174	161	124	75	81	44